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Power input, soft-latch, supplies

File: power.kicad_sch

3-phase power stage and current sensing

File: bridge.kicad_sch

STM32G431 controller

File: mcu.kicad_sch

Connectors and I/O conditioning

File: io.kicad_sch

DESIGN SUMMARY

– Battery: 48 V nominal (13S Li-ion, 54.6 V full). Abs. max 60 V (TVS SMCJ60CA, 100 V FETs/drivers/bucks).

– Battery current limit 35 A (firmware, from FOC power estimate). Phase current limit 60 A (firmware), hard trip 80 A.

– Power stage: 6 x IPT015N10N5 (100 V, 1.5 mOhm, TOLL) driven by 3 x LM5109B half-bridge drivers @ 20 kHz, 4.7 Ohm gate resistors.

– Current sensing: in-line 0.5 mOhm 4-terminal (Kelvin) phase shunts + INA240A1 (G=20, mid-rail ref) -> +/-165 A full scale.

– Control: STM32G431CB @170 MHz, hall-sensor FOC with angle interpolation; hall/R/L auto-detect over UART.

– Supplies: LM5164 48->12 V (gate drive), LM5164 48->5 V (halls/throttle), AP2112 5->3.3 V (logic).

– Soft power: momentary button to GND wakes the bucks via Q1, MCU latches with PWR_HOLD; long press = off.

– I/O: hall + motor NTC (JST-PH6), throttle (PH3), FWD/REV switch (PH2), brake (PH2), power button (PH2), UART 115200 8N1 3.3 V (PH4: 3V3, TX, RX, GND), SWD (1x5 header).

LAYOUT RULES: 2 oz copper min (4 layer preferred), bus/phase pours >= 10 mm wide, FET tabs on thermal-via arrays to an aluminium heat spreader, Kelvin-route INA240 inputs from the shunt pads, DC-link ceramics at each half bridge.

Top level

STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3

48V 35A hub motor controller

Sheet: /

File: bldc48.kicad_sch

Title: 48V 35A BLDC hub motor controller

Size: A3

Date: 2026-10-06

Rev: A

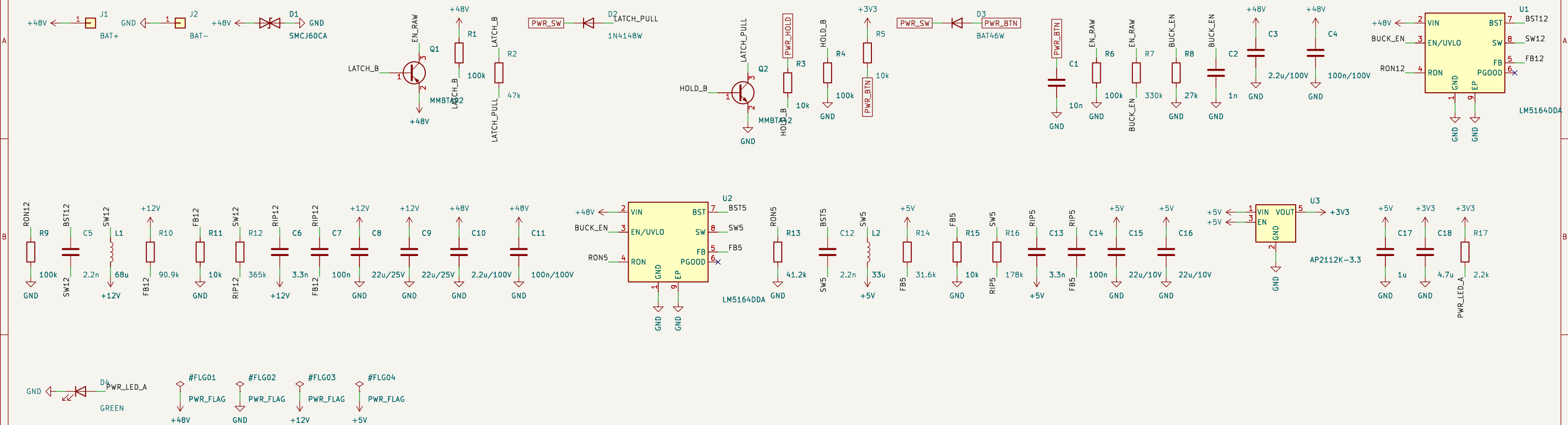
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Power input, soft-latch, supplies

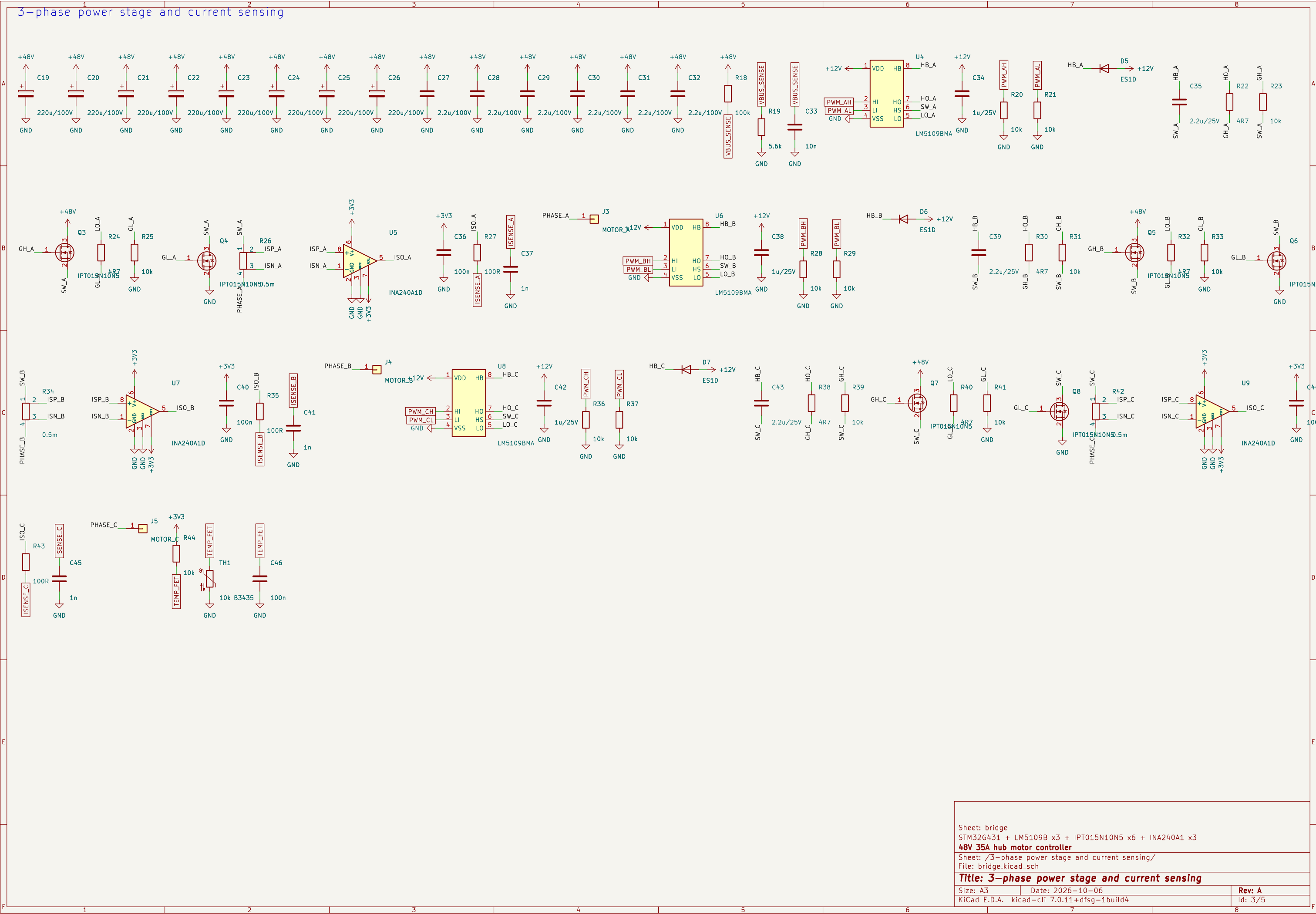


Sheet: power
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3
48V 35A hub motor controller
Sheet: /Power input, soft-latch, supplies/
File: power.kicad_sch

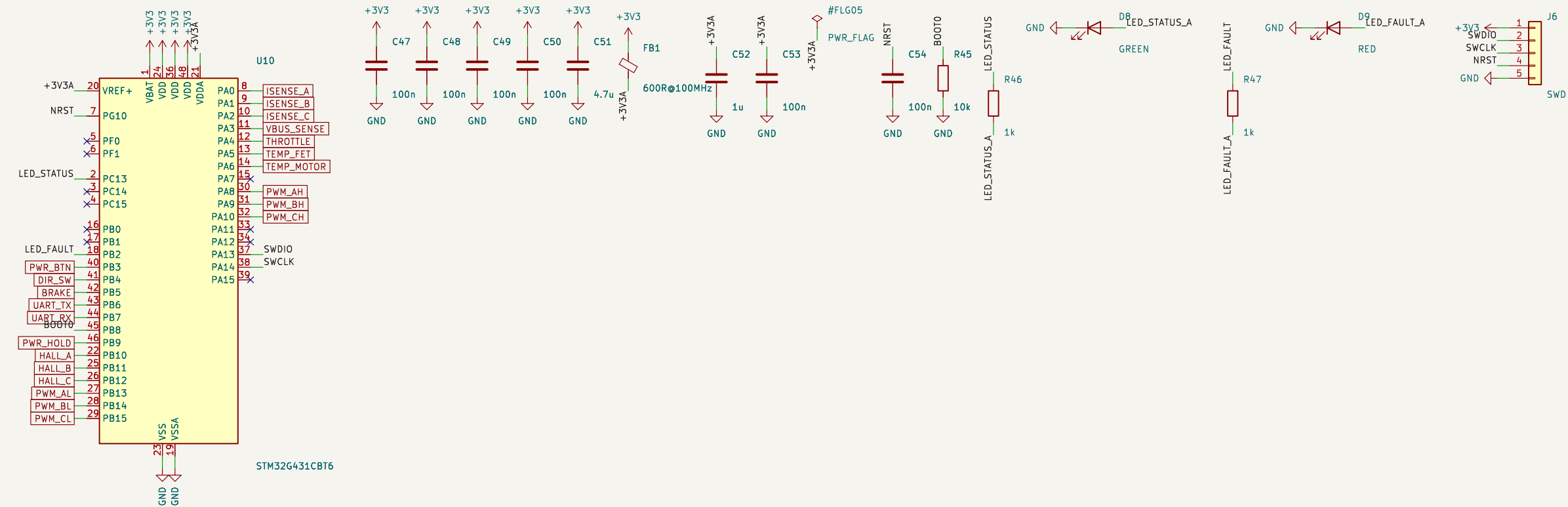
Title: Power input, soft-latch, supplies

Size: A3	Date: 2026-10-06	Rev: A
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3-phase power stage and current sensing

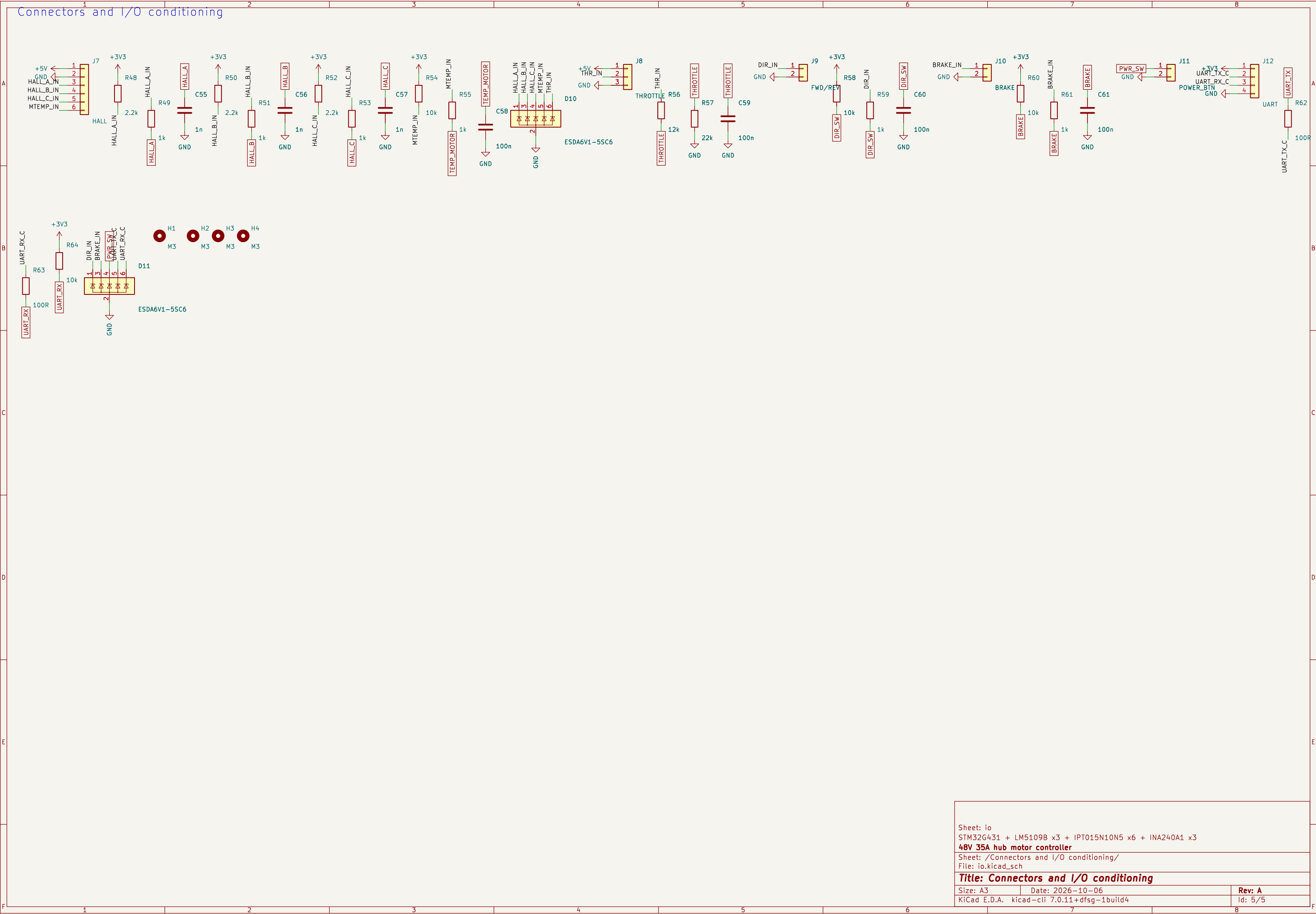


STM32G431¹ controller



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Sheet: mcu
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3
48V 35A hub motor controller
Sheet: /STM32G431_controller/
File: mcu.kicad_sch
Title: STM32G431 controller
Size: A3 | Date: 2026-10-06
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
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Connectors and I/O conditioning



Sheet: io
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3
48V 35A hub motor controller
Sheet: /Connectors and I/O conditioning/
File: io.kicad_sch

Title: Connectors and I/O conditioning

Size: A3	Date: 2026-10-06	Rev: A
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